AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

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(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I186*

Team Name: *NEXUS*

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Problem Statement

During emergencies, patients are often directed to the nearest hospital without considering its current capacity, available facilities, required treatment and travel conditions. This can lead to overcrowding and delays in receiving appropriate care. How can an intelligent system dynamically match emergency patients with the most suitable available hospital based on medical requirements, hospital capacity and estimated travel time?

Problem Understanding

- Who experiences this problem?
  Emergency patients, their families, ambulance crews, and overcrowded hospitals all suffer when patients are sent to the nearest hospital without regard to its capacity, facilities, or traffic. The result is delayed care and, in time-critical cases, worse outcomes.

- What causes the problem?
  Patients are sent to the nearest hospital because there is no live data on beds, specialists, or traffic, and nothing matches the patient's medical needs to a hospital's capabilities. Hospitals and ambulances also work in separate systems, so crews decide with little information.

- How does it affect users or stakeholders?
  Patients face delayed treatment and worse outcomes, and their families face added stress. Ambulance crews lose time when hospitals turn them away, and overcrowded hospitals deal with overloaded staff and strained resources while nearby hospitals sit underused.

Existing Gaps & Challenges

- Limitation 1: No real-time data sharing. Hospital bed and specialist availability is updated manually or only periodically, so ambulance crews and existing systems decide with outdated information and can still send patients to a hospital that is already full.
- Limitation 2: Medical needs are not matched to hospital capabilities. Most existing systems pick a hospital by distance alone, ignoring the patient's condition, so a stroke or trauma patient can be taken to a hospital without the right specialist or equipment.
- Limitation 3: Live travel conditions are ignored. Existing systems rely on fixed distance and do not account for live traffic, road closures, or changing travel times, so the hospital that looks nearest on the map may not be the fastest to reach.

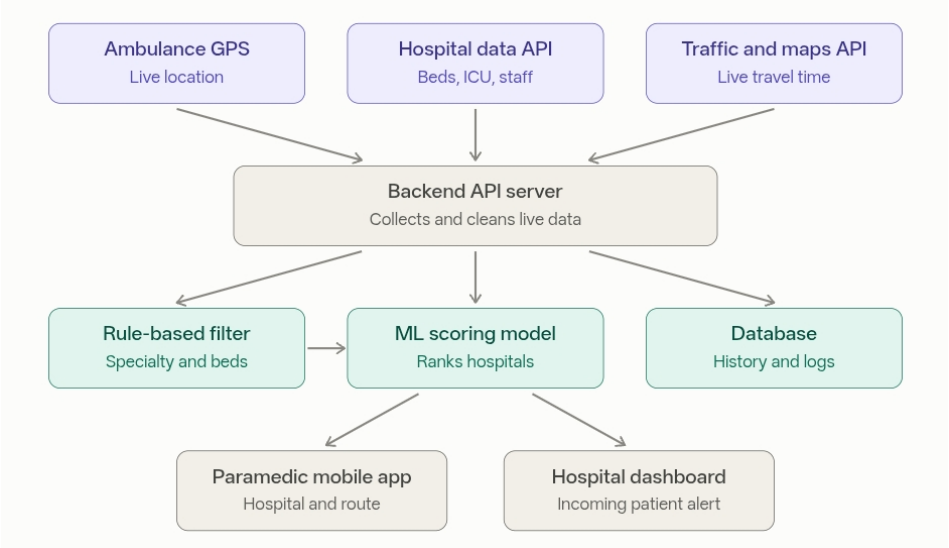
Our Proposed Solution

- Core idea: An intelligent emergency hospital-matching system that, instead of sending patients to the nearest hospital, picks the best-suited one in real time. It scores every hospital on the patient's medical needs, live bed and specialist availability, and live travel time, then recommends the top choice to the ambulance crew and alerts that hospital in advance. It keeps re-checking during the trip and re-routes if conditions change.

- How the solution works at a high level?
  The paramedic enters the patient's condition, and the system checks live hospital capacity, specialties, and traffic. It ranks hospitals by medical match, availability, and travel time, sends the ambulance to the best one, alerts that hospital in advance, and re-routes if conditions change.

- Main value delivered to users: Patients get faster, more suitable care, ambulance crews get a clear and reliable destination, and hospitals get balanced patient loads and advance notice, which together reduce delays, overcrowding, and risk to life.

Technical Approach



The paramedic app sends the patient's condition to the backend, which pulls live hospital and traffic data, filters and ranks hospitals using the ML model, and sends the best hospital and route back to the app while alerting that hospital's dashboard.

Tech Stack

Frontend: Paramedic mobile app

Backend: Python

Database: Firebase for real time updates

AI/ML, APIs, or Hardware: scikit-learn, Google maps API, Ambulance GPS

Innovation & Uniqueness

- What is new or different?
  Unlike systems that send patients to the nearest hospital, ours combines the patient's medical needs, live hospital capacity, and real-time travel time in one decision, and re-routes if conditions change during the trip.

- Why is this approach suited to the problem?
  Because the problem comes from sending patients to the nearest hospital without live information, our approach fixes exactly that by using real-time capacity, medical needs, and travel time to pick the right hospital and adjusting if conditions change, which reduces delays and overcrowding.

- What is the key differentiator?
  Existing systems choose a hospital by distance alone, while ours chooses by best fit, combining medical needs, live capacity, and travel time.

Feasibility & Scalability

- **Feasibility:** The system can be built with existing, widely available tools such as mobile and web frameworks, cloud databases, and mapping APIs for live traffic, so no new hardware is needed. Hospitals already maintain bed and specialty records, so connecting them through a simple API or dashboard is practical, and a prototype can be built and tested quickly with simulated data before any real-world pilot.
- **Scalability:** The system can start with a few hospitals in one city and expand to more hospitals, cities, and regions, because it runs on cloud infrastructure that grows with demand. New hospitals only need to connect to the data API, and the same matching logic works anywhere, with the ML model improving as more data is collected.
- **Risks:** The main risks are outdated or incomplete hospital data, hospitals being reluctant to share live information, patient data privacy, and system or internet failure during an emergency. These can be reduced by automatic data updates, secure and encrypted data handling, a fallback to nearest-hospital suggestions if the system goes offline, and keeping the final decision with the paramedic.

Cost for Product (Optional)

- Development or hardware cost: 3-5 lakh for development
- Hosting or maintenance cost: 5,000-15,000 per month for hosting and 30,000-75,000 for maintenance
- Estimated total: 4-7 lakh INR

Use Cases

- Use case 1: Use case 1: Heart attack patient. A paramedic reaches a patient with chest pain and enters the condition into the app. Instead of going to the nearest hospital, which has no free cardiac ICU bed, the system finds a slightly farther hospital with a cardiologist and an open bed, sends the ambulance there, and alerts the hospital so the cardiac team is ready on arrival.
- Use case 2: : Road accident with severe injuries. An ambulance picks up a trauma patient after a highway accident. The nearest hospital is overcrowded and has no trauma surgeon on duty, and the usual route is jammed. The system picks a hospital with an available trauma team and a faster route, then alerts it in advance so the operating room is ready when the patient arrives.
- Use case 3: Pregnancy emergency. A pregnant woman in labor with complications is picked up by an ambulance. The nearest hospital has no neonatal ICU, so the system selects a hospital with an obstetrician and a free neonatal bed, and alerts it so the maternity and newborn care teams are prepared when she arrives.

Impacts and Benefits

- Patients get faster and more suitable treatment, which improves their chances of survival and recovery. Ambulance crews get a clear, reliable destination without wasted trips, and hospitals get balanced patient loads and advance notice to prepare. Families also face less stress and uncertainty during emergencies.

- Socially, it saves lives by getting patients to the right care faster. Economically, it cuts wasted ambulance trips and uses hospital resources more efficiently. Environmentally, shorter routes mean less fuel use and fewer emissions.

- Expected outcomes include shorter time from pickup to treatment, fewer patients transferred between hospitals after arrival, lower overcrowding in emergency departments, and better use of hospital beds across the city. These can be measured by tracking average response and treatment time, the transfer rate, and hospital occupancy levels before and after the system is introduced.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 3	Target 3.8	Good Health and Wellbeing	It improves access to timely, quality emergency care by matching each patient to a hospital that has the right facilities and capacities.
SDG 9	Target 9.1	Industry, Innovation and Infrastructure	Uses real time data and intelligent technology to make existing healthcare infrastructure more efficient and reliable.

Individual Member Contribution

Member	Contribution or responsibility
Dheebesh L N	Research through different papers and content for chart, Topic selection, came up with Problem Statement
Jayavanth J	Research through different papers, simulated different use cases, chartwork
Akash T Anish	Content Research for Presentation and Chart, Chartwork
Radhakrishnan Ganesan	Content research for Presentation, Created Presentation, Chartwork

Yukti ID

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